

Reply to reports on article JBES-P-2022-0567.R1

First of all, we would like to thank the editor, the associate editor and the reviewers for their many comments and suggestions which were very helpful to improve the paper. In the revision, we have addressed all comments of the referees and have rewritten the paper accordingly. Please find our point-by-point responses to the referees below. In our responses, we have indicated where alterations to the manuscript have been made. Please also note that we have tried to add as much of the new material as possible to the main body of the paper. However, to meet the 35-pages-limit, we still had to relegate parts of the material. If required, we can of course rearrange the material, shifting part of the simulations to the main body of the text while moving other parts to the Supplement.

Reply to the associate editor

1. *My apologies, but I maintain that your H_0 (page 3) is unnatural and uninteresting. In the trend analysis of multiple time series, it is extremely unlikely that the natural ‘position of ignorance’ is that all trend curves are the same. What if, for example, I already know or suspect that the curves form distinct groups? It is then something of a wasted effort to start from the position that they do not differ at all. Also, the null hypothesis is not robust to high-dimensionality as it is more likely that some curves will differ as I observe a higher number of variates (I understand that you do not do this in the paper but this only makes it even less interesting, in my view). As a cartoon illustration of the first point above, if I suspect that $p - 1$ (say) curves may be the same, but 1 is distinctly different, do I need to first remove the offending one before performing your test, in which case I enter the territory of post-selection inference (which you do not cover), or do I not need to remove the offending one, in which case I am pretending that my H_0 is different from what it is in reality?*

We appreciate your perspective regarding the null hypothesis H_0 and acknowledge that in many real-world applications, assuming all trends to be identical may seem restrictive. However, we argue that our choice of H_0 is methodologically justified because it represents the simplest, most parsimonious starting point for testing. This aligns with the traditional Neyman-Pearson framework where the null serves as a benchmark for detecting departures. Furthermore, our multiscale method enhances this approach by not only rejecting H_0 but also identifying specific intervals and pairs of series where differences occur, providing actionable insights beyond a binary rejection decision.

We also understand your concern about the robustness of our test to high di-

dimensionality of the covariates. However, while our paper does not explicitly address high-dimensional asymptotics, the core methodology could be extended to incorporate sparsity-driven approaches in the future.

Regarding the cartoon illustration, suppose it is indeed true that 1 curve is distinctly different from the other $p - 1$ curves. First of all, in this case, the rest $p - 1$ may be different among themselves with some having more in common with the first one, and the others exhibiting much more pronounced dissimilarities from it. Second, even if all $p - 1$ curves are exactly the same, it very well may be that the difference between them and the first curve occurs only on some subinterval of $[0, 1]$. In both these cases, our test would be able to pinpoint this difference, and the subsequent clustering will not necessarily result in a separate group for the first, offending curve. Hence, even in this cartoon example, we would still advise applying the statistical test for all the p curves at the same time.

2. *The clustering part is nice and perhaps it could form the main thread of a much shorter paper.*

We have taken your suggestion into account and put more focus on the clustering part in the current paper. (Marina: I have not done this.)

3. *The empirical application is interesting, but this is thanks to the use of clustering rather than as a consequence of testing - the null is obviously inappropriate, as Figure 6 suggests, so there is no point in assuming it only to surely reject it. In other words, “let us test our null of equality between all trends” doesn’t seem a scientifically valid suggestion for this dataset. The suggestion of clustering the countries is fine, though.*

We appreciate your point of view about the empirical application, and we would like to stress once more that the null of equality between the trends is just a starting point of our analysis. Many econometric procedures, including cointegration tests and structural break tests, involve null hypotheses that are often rejected but remain essential for empirical analysis.

We acknowledge that in this particular example clustering may be more interesting than simply testing the global null. However, it is important to emphasize that our testing framework is not just a preliminary step but an integral part of the analysis. Specifically, our clustering method is directly informed by the results of the multiscale test: the distance measure that we use for applying the hierarchical clustering is exactly the pairwise test statistics that was used in computation of the global test statistics.

We would also like to reiterate that our method can identify cases where trends

are similar over certain time intervals even if they differ globally. This is a nuanced insight that clustering alone cannot provide. For example, our test does not detect any significant differences between Belgium and any other country except Netherlands in the second half of the observed interval, a statement that can not be made neither from the visual inspection of Figure 6, nor from the clustering results alone.

4. *Can your model (1.1) include autoregression? Either way, it would be good to see it spelled out explicitly.*

While the current formulation does not explicitly include autoregressive terms (due to the restriction on the relationship between the covariates and the errors introduced in Assumption (C7)), autoregressive dynamics can be incorporated through the error term ε_{it} as, for example, $\varepsilon_{it} = a\varepsilon_{i(t-1)} + \eta_{it}$, where η_{it} are i.i.d. normal random variables with zero mean and finite variance and $|a| < 1$ ensures that Assumption (C3) is satisfied. We mention the autoregressive processes briefly on p. 7 and in Remark 2.1 on p. 8-9. Moreover, the size and power simulations are performed including the autoregressive structure in the error terms (see p. 1 in the Supplement). (Marina: should we say it explicitly somewhere in the text?)

5. *I was unconvinced by the subtraction of the overall level on p. 6. Firstly, this clearly risks eliminating useful information (what if some curves differ from others in the overall level)? Secondly, I am not sure if it is meaningful to look at averages of (e.g. increasing) trends - for example, what is the interpretation of a “long-term average house price level in [a] country”? In “Time Series 101”, we teach students not to take averages of non-stationary time series!*

Your concerns about the potential loss of meaningful information and the interpretation of averages of non-stationary time series are well taken. However, the subtraction of the overall level in our model serves a specific identification purpose. Formally, we impose the normalization condition $\int_0^1 m_i(u)du = 0 \forall i$, which effectively centers each trend around zero. Without this constraint, the model suffers from non-identifiability because the additive fixed effects α_i and the trend functions m_i would not be uniquely separable.

Furthermore, our methodology is designed to detect dynamic differences in trend behavior over time, rather than static level differences. The subtraction of the overall mean ensures that the test focuses on structural variations (e.g., changes in slope, curvature) rather than trivial shifts in baseline levels.

As for taking the averages of non-stationary time series, we agree that it is not an advisable practice in general. However, in our case non-stationarity in

$\{Y_{it} : 1 \leq t \leq T\}$ for each $i = 1, \dots, n$ is introduced through the deterministic trend function m_i , not through the random covariates $\{\mathbf{X}_{it} : 1 \leq t \leq T\}$ or the error process $\mathcal{E}_i$ (see Assumptions (C4) and (C1), respectively). Since m_i is a deterministic smooth function, this operation does not suffer from the same issues as averaging non-stationary stochastic processes.

6. *(C5) - you use the delta notation before defining it. Why do your covariates X have to be random, e.g. why can't you have one of them as (say) a linear trend?*

We appreciate the Associate Editor's careful reading and pointing out the use of the delta notation before its formal definition in assumption (C5). We have now revised the manuscript to ensure that all notations are introduced prior to their use, improving the clarity of the presentation.

Regarding the requirement that the covariates $\mathbf{X}_{it}$ must be random rather than allowing for a deterministic covariate, this assumption is primarily motivated by the identification reasons within our theoretical framework. Specifically, if we include a linear (or any other smooth) trend as one of the covariates, it would create ambiguity about whether the observed time variation arises from the trend function m_i or it is the effect of this covariate. Such ambiguity leads to non-identifiability of the deterministic trend function. Random covariates provide exogenous variation that helps disentangle their effects from those of the nonparametric trend m_i . This stochastic variation acts as a natural source of identification which is essential in our approach.

7. *Your comment (iii) on p. 9. But isn't it the case that for near-unit root processes, you will have a variety of different scenarios (some stochastic trends pointing strongly upwards, some downwards, some with no specific direction) - a situation that should be easy to discriminate from your typical scenarios of interest?*
8. *P. 11. "Throughout the paper, we assume that the long-run error variance is the same for all i , (...) [but we] use a different estimator of σ^2 for each i - this seems suboptimal.*

(Marina: Should we rerun everything with the averages of the long-run error variances? I think I tried this and the results were somewhat similar.)

9. *I was disappointed to see that the method was not able to handle missing values.*

While the current version of our method does not explicitly handle missing data, we recognize that this is a valuable direction for further development. The main challenge in handling missing values arises from the dependence of our multi-scale test on kernel-based local averages. Specifically, our methodology assumes equally spaced observations to ensure the consistent estimation of nonparametric

trend functions. We believe that it is possible to modify the multiscale framework to handle irregularly spaced observations in the future research. For example, the kernel weights in the local linear estimators could be adapted to reflect the actual spacing between observed points, similar to techniques used in nonparametric regression with irregularly spaced data.

Reply to referee 1

Thank you very much for the careful reading of our manuscript and the interesting suggestions. In our revision, we have addressed all your comments. Please see our replies to them below.

1. *To address some “big picture” concerns about practical relevance, the authors clarify that rather than the null hypothesis of “no differences ever”, the main focus is on the alternative hypotheses, and specifically the ability to detect 1) which series are different and 2) when. This is a reasonable response. However, the simulation studies do not really showcase these priorities. First, simulation results do not appear at all in the main paper. If there is a question about practical relevance of the methods (which the review team did raise), then empirical support should be elevated above some of the technical details. Second, the global size and power tables do not clearly illustrate such local testing capabilities—these metrics would require a much larger collection of data-generating processes to measure performance across a large spectrum of alternative hypotheses. Third, the suggested competitors have not been included, and no compelling reasons were given (more below). Finally, the authors have kept $m_i = 0$ to simulate data under the null “WLOG”. As I noted before, this is not WLOG for all methods; that it *is* WLOG for their method suggests that they may observe comparative gains over other approaches for smaller sample sizes (n).*
2. *I do not understand why the authors are resistant to add the simple competitors suggested (additive models w/ CI-based testing and Bayesian analogs). Their reasoning was 1) these methods do not offer the same theory guarantees and 2) they include another method, sizer, and a simplified version of their approach. For 1): if the theory actually offers practical gains, then this should show up clearly empirically and further advertise the proposed methods! For 2): sizer is a reasonable competitor (for $n=2$), but I don’t see why that would eliminate consideration of any alternatives. If the question is “which curves are different and when”, a natural first approach would be to fit a nonparametric model to each curve and see where/when the confidence/credible intervals do not overlap (w/ or w/o a Bonferroni correction). Beating this approach will be more convincing for general readers.*

Kim (2016)

3. *Stochastic volatility (SV) is widely important in the types of applications that the authors mention. Yet the authors 1) claim that extensions to SV would be challenging, 2) elect not to consider any simulation designs with SV to examine robustness, and 3) claim that SV can be handled with log-transformations (and then using the same mean models as proposed). Perhaps 1) is true, but 2) is important based on the claims in the paper and the relevance for this journal. Finally, 3) is not correct for the most widely-used SV models.*

Reply to referee 2

Thank you very much for the careful reading of our manuscript and the final acceptance. We have addressed your remaining question in the revision. Please see our reply to it below.

1. *Thank you very much for your very careful revision. I have just one remaining "cosmetic" question: can you explain why in the third row of your figures such as Figure 1, page 27, we see several intervals one on top of the other? In other words is there a meaning of the y-axis of this lower plots, what is its scale? Thank you in advance for adding this information (which I might have missed elsewhere?).*

The vertical positioning of the intervals from the sets $\mathcal{S}^{[i,j]}(\alpha)$ does not bear any additional information and is used for better graphical representation only. We have added this information to the note under Figures 1-4, and we apologize for not making it clear beforehand.

References

KIM, K. H. (2016). Inference of the trend in a partially linear model with locally stationary regressors. *Econometric Reviews*, **35** 1194–1220.